

Maths – Assignment-1 For Biocon Academy AI Course

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(Last date for submission : on or before 5 pm, April 24, 2026)

Figures should be plotted in R and screen shots should be attached, where required.

1. Identify the linear and nonlinear functions among the following:

(i) $f(x) = 3.1x - 5.6$

(ii) $f(x) = 5x + 3x + 9$

(iii) $g(x) = 3x(x + 3)$

(iv) $g(x) = \frac{3x + x}{2x}$

2. Find the slope and intercept of the following straight lines:

(i) $y = 5x - 0.5$

(ii) $x(5 + \frac{6}{x})$

(iii) $3x - 0.2x + 12$

(ii) $xy = x^2 + 5x$

3. Identify the functions and equations among the following:

(i) $y = 3x + 5$

(ii) $3x + 2 = 5$

(iii) $5^2 = 25$

(iv) $y^2 = 4x$ (Is this a function?)

4. Plot the following functions in the given range using R plots:

(i) $f(x) = 5x + 3$ and $f(x) = -5x + 3$ on same plot in the range $[-5, 5]$ in steps of 0.2

(ii) $y = x^2$ and $y = 2x^2$ in the same plot in the range $[-3, 3]$ in steps of 0.2

(iii) $y = 8x - x^2 - x^3$ in the range $[-4, 4]$ in steps of 0.2